

HDOC DAY-26 ASSIGNMENT

Menu-Driven Right-Aligned Patterns Using Switch-Case

1. Problem Understanding

The task is to write a menu-driven C program using switch-case to print three right-aligned patterns. A separate parameterless user-defined function with void return type is used for each menu option. Each function reads the number of rows, uses nested loops without arrays, and displays the selected pattern.

- Print a right-aligned increasing triangle using asterisks.
- Print a right-aligned increasing triangle of natural numbers.
- Print a right-aligned decreasing triangle using asterisks.

2. Standard Test Cases

Test Case	Choice	Input	Expected Output	Status
TC1	1	4	Right-aligned increasing star triangle	Pass
TC2	1	3	Right-aligned increasing star triangle	Pass
TC3	2	4	Right-aligned increasing number triangle	Pass
TC4	2	3	Right-aligned increasing number triangle	Pass
TC5	3	4	Right-aligned decreasing star triangle	Pass
TC6	3	3	Right-aligned decreasing star triangle	Pass

3. Algorithm

A. Common Menu Logic

1. Start.
2. Display the menu options.
3. Read the user's choice.
4. Use switch-case to select the required pattern.
5. Call the corresponding void function.
6. The selected function reads the number of rows n and prints the pattern using nested loops.
7. If the choice is invalid, display "Invalid choice".
8. Stop.

B. Right-Aligned Increasing Triangle

1. Start.
2. Read the number of rows n .
3. For each row i from 1 to n , print $n - i$ leading spaces.
4. Print i asterisks in the same row.
5. Move to the next line after each row.

6. Stop.

C. Right-Aligned Increasing Number Triangle

1. Start.
2. Read the number of rows n .
3. For each row i from 1 to n , print $n - i$ leading spaces.
4. Print natural numbers from 1 to i in the same row.
5. Move to the next line after each row.
6. Stop.

D. Right-Aligned Decreasing Triangle

1. Start.
2. Read the number of rows n .
3. For each row i from n down to 1, print $n - i$ leading spaces.
4. Print i asterisks in the same row.
5. Move to the next line after each row.
6. Stop.

4. C Program

```
#include <stdio.h>
void increasingStar()
{
    int n, i, j;
    printf("Enter the number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= i; j++)
            printf("* ");
        printf("\n");
    }
}
void increasingNumber()
{
    int n, i, j;
    printf("Enter the number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= i; j++)
            printf("%d ", j);
        printf("\n");
    }
}
```

```

}
void decreasingStar()
{
    int n, i, j;
    printf("Enter the number of rows: ");
    scanf("%d", &n);
    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= i; j++)
            printf("* ");
        printf("\n");
    }
}
int main()
{
    int choice;
    printf("MENU\n");
    printf("1. Right-aligned increasing triangle\n");
    printf("2. Right-aligned increasing number triangle\n");
    printf("3. Right-aligned decreasing triangle\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    switch (choice)
    {
        case 1:
            increasingStar();
            break;
        case 2:
            increasingNumber();
            break;
        case 3:
            decreasingStar();
            break;
        default:
            printf("Invalid choice\n");
    }
    return 0;
}

```

5. Compilation and Execution Evidence

```
kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)-[~/Desktop]
└─$ gedit question.c

** (gedit:2252): WARNING **: 09:59:46.978: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:2252): WARNING **: 09:59:46.978: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found

(kali@kali)-[~/Desktop]
└─$ gcc question.c -o question

(kali@kali)-[~/Desktop]
└─$ ./question
MENU
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
Enter your choice: 1
Enter the number of rows: 4
  *
 * *
* * *
* * * *

(kali@kali)-[~/Desktop]
└─$ ./question
MENU
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
Enter your choice: 1
Enter the number of rows: 3
  *
 * *
* * *
```

```
kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)-[~/Desktop]
└─$ ./question
MENU
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
Enter your choice: 2
Enter the number of rows: 4
  1
 1 2
1 2 3
1 2 3 4

(kali@kali)-[~/Desktop]
└─$ ./question
MENU
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
Enter your choice: 2
Enter the number of rows: 3
  1
 1 2
1 2 3

(kali@kali)-[~/Desktop]
└─$ ./question
MENU
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
Enter your choice: 3
Enter the number of rows: 4
* * * *
* * *
* *
*
```

```
kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
Enter your choice: 3
Enter the number of rows: 4
* * *
* *
*
*

(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
Enter your choice: 3
Enter the number of rows: 3
* *
*
*

(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
Enter your choice: 4
Invalid choice

(kali@kali)-[~/Desktop]
$ 4
4: command not found

(kali@kali)-[~/Desktop]
```

6. Verification of All Test Cases

Test Case	Choice	Input	Expected Result	Actual Result	Status
TC1	1	4	Increasing star triangle	Increasing star triangle	Pass
TC2	1	3	Increasing star triangle	Increasing star triangle	Pass
TC3	2	4	Increasing number triangle	Increasing number triangle	Pass
TC4	2	3	Increasing number triangle	Increasing number triangle	Pass
TC5	3	4	Decreasing star triangle	Decreasing star triangle	Pass
TC6	3	3	Decreasing star triangle	Decreasing star triangle	Pass
